

Disturbance Observer-based Nonsingular Fast Terminal Sliding Mode Fault Tolerant Control of a Quadrotor UAV with External Disturbances and Actuator Faults

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Abstract: This paper addresses a nonsingular fast terminal sliding mode fault tolerant control (NFTSMFTC) scheme based on disturbance observer (DO) for a quadrotor UAV in the presence of external disturbances and actuator additive faults. Firstly, by analyzing the dynamic characteristics of a quadrotor UAV, the model is decoupled into outer-loop (position subsystem) and inner-loop (attitude subsystem). Secondly, the DO is utilized to deal with external disturbances and actuator additive faults effectively. Furthermore, nonsingular fast terminal sliding mode control (NFTSMC) can avoid the chattering problems of traditional sliding mode control (SMC) and singularity problems of terminal SMC. Moreover, compared with nonsingular terminal SMC, it behaves the faster convergence rate. So the NFTSMC method is applied to design controller. Finally, in the framework of Lyapunov theory, the stability of the closed-loop system is analyzed, and the tracking error is asymptotically stable. The effectiveness of the proposed control scheme is verified by simulation results.

Keywords: Actuator additive fault, disturbance observer (DO), external disturbance, nonsingular fast terminal sliding mode (NFTSM), quadrotor UAV.

1. INTRODUCTION

The quadrotor UAV is a rotor-type autonomous vehicle which can take off and land vertically in small space. Due to its advantages of light weight, small structure, simple and easy operation, it is widely used in military and civilian fields, such as rescue, surveillance, aerial photography, agricultural reconnaissance [1–4]. However, controlling quadrotor UAV is a challenging task on account of the nonlinear, strong coupling, under-actuated, susceptibility to external disturbances, uncertainties and other factors.

Many linear control strategies were successively applied to the quadrotor UAV, such as linear quadratic regulator (LQR) [5], trajectory linearization control (TLC) [6], PID control [7], and H_∞ control [8]. Although linear control approaches have been widely used, they can't satisfy specific conditions for nonlinearity systems. To overcome this limitations, nonlinear control strategies have been used for controller design of a quadrotor UAV which include backstepping control [9–11], SMC [12–15], etc.

In [9], an adaptive backstepping fault-tolerant controller was designed by using the adaptive law to estimate ac-

tuator additive fault. By combining with adaptive interval type-2 fuzzy logic system, a robust backstepping controller was designed in [10] for a quadrotor UAV under actuator faults. Based on DO, a backstepping control strategy was proposed in [11] to achieve the stable tracking of a quadrotor UAV with parameter uncertainty. In [12], based on SMC, both passive and active fault-tolerant control laws were designed. A nonlinear robust adaptive fault tolerant tracking controller of a quadrotor UAV was designed in [13] under actuator faults. Even though the controllers in [9–13] can achieve the good tracking performance, they only considered the actuator faults without external disturbances. For the attitude tracking control problem of a quadrotor UAV under actuator faults and external disturbances, a DO-based SMC strategy was designed in [14]. The attitude control problem of a quadrotor UAV with time-varying mass was investigated in [15]. However, the position tracking problem was not considered in [14,15]. Furthermore, a super twisting sliding mode controller was designed by utilising a cascaded inner outer loop structure in [16]. In [17], a fractional order sliding mode controller based on backstepping method was de-

Manuscript received October 16, 2020; revised May 18, 2021; accepted June 8, 2021. Recommended by Associate Editor Nam H. Jo under the direction of Editor PooGyeon Park. This research was supported by Natural Science Foundation of Hebei Province (F2020203105), National Natural Science Foundation of China (61503323).

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signed under external disturbances and variations in load. A controller based on the combination of adaptive neural network and SMC was proposed for a quadrotor UAV with parameter uncertainty and external disturbances [18]. A self-tuning proportional integral regulator-based robust adaptive controller for a quadrotor UAV was designed in [19]. Nevertheless, the above results [17–19] didn't comprehensively consider external disturbances and actuator faults. In [20], a continuous multivariable super-twisting-like control scheme was proposed for a quadrotor UAV, but the position control problem was not considered. In [21], a nonlinear robust adaptive SMC scheme was proposed for miniature UAV with uncertainty. In [22], the adaptive law was used to deal with the external disturbance, based on the combination of backstepping technique and fast terminal SMC, a robust controller was proposed for a quadrotor UAV. However, the traditional SMC [21] and fast terminal SMC [22] respectively lead to the inevitable chattering problem and singularity problem.

Motivated by above discussions, in this paper, the position and attitude tracking control problems of a quadrotor UAV are simultaneously studied, where the external disturbance and actuator faults are considered. To facilitate controller design, the complex quadrotor UAV model is decomposed into inner-loop attitude subsystem and outer-loop position subsystem. The DO is used to deal with external disturbances and actuator faults. The main contributions and innovations are summarized as follows:

First, a NFTSMC scheme is proposed, which avoids the chattering problem of traditional SMC method in [21] and the singularity problem of FTSMC method in [22].

Second, in [9–13], external disturbance on a quadrotor UAV was ignored. The position tracking control problem was not considered in [14,15]. In this paper, the position and attitude tracking control problems were simultaneously considered, where the influences of actuator faults and external disturbances are taken into account.

Third, in [23], the DO was used to deal with external disturbances, and adaptive law was constructed to estimate the actuator fault. Whereas in this paper, DO is employed to handle external disturbances and actuator faults, which simplifies the design process of controller.

The structure of this paper is organized as follows: The quadrotor UAV model and the control goal are described in Section 2. In Section 3, the quadrotor UAV system is divided into attitude subsystem and position subsystem. The corresponding controller is designed. In Section 4, simulation and analysis are given. Finally, conclusion is given in Section 5.

2. THE MODEL OF A QUADROTOR UAV

The model of a quadrotor UAV as follows [24,25]:

$$\ddot{x} = \frac{1}{m}(\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi)u_1 - \frac{g_1 \dot{x}}{m},$$

$$\begin{aligned} \ddot{y} &= \frac{1}{m}(\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi)u_1 - \frac{g_2 \dot{y}}{m}, \\ \ddot{z} &= \frac{1}{m}(\cos \phi \cos \theta)u_1 - g - \frac{g_3 \dot{z}}{m}, \\ \ddot{\phi} &= \frac{a}{J_{xx}}u_2 - \frac{g_4 a}{J_{xx}}\dot{\phi}, \\ \ddot{\theta} &= \frac{a}{J_{yy}}u_3 - \frac{g_5 a}{J_{yy}}\dot{\theta}, \\ \ddot{\psi} &= \frac{a}{J_{zz}}u_4 - \frac{g_6 a}{J_{zz}}\dot{\psi}, \end{aligned} \quad (1)$$

where x, y, z and ϕ, θ, ψ are the position and attitude of quadrotor UAV, respectively. m is the mass, g denotes the gravity acceleration, a is the wing length, J_{xx}, J_{yy}, J_{zz} respectively represent the inertias on the x, y, z axis. $g_i (i = 1, 2, \dots, 6)$ is the drag coefficients. u_1, u_2, u_3, u_4 are the four control variables.

Considering external disturbances and actuator additive faults, system (1) can be rewritten and divided into attitude subsystem (2) and position subsystem (3).

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= a_1 x_2 + b_1 u_2 + D_1, \\ \dot{x}_3 &= x_4, \\ \dot{x}_4 &= a_2 x_4 + b_2 u_3 + D_2, \\ \dot{x}_5 &= x_6, \\ \dot{x}_6 &= a_3 x_6 + b_3 u_4 + D_3, \\ \dot{x}_7 &= x_8, \\ \dot{x}_8 &= a_4 x_8 + \frac{\cos x_1 \cos x_3}{m} u_1 - g + D_4, \\ \dot{x}_9 &= x_{10}, \\ \dot{x}_{10} &= a_5 x_{10} + \frac{u_x}{m} u_1 + D_5, \\ \dot{x}_{11} &= x_{12}, \\ \dot{x}_{12} &= a_6 x_{12} + \frac{u_y}{m} u_1 + D_6, \end{aligned} \quad (2)$$

where $x = (x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}) = (\phi, \dot{\phi}, \theta, \dot{\theta}, \psi, \dot{\psi}, z, \dot{z}, x, \dot{x}, y, \dot{y})$, $a_1 = \frac{-g_4 a}{J_{xx}}$, $a_2 = \frac{-g_5 a}{J_{yy}}$, $a_3 = \frac{-g_6 a}{J_{zz}}$, $a_4 = \frac{-g_3}{m}$, $a_5 = \frac{-g_1}{m}$, $a_6 = \frac{-g_2}{m}$, $D_i = b_i f_i + d_i$, $b_1 = \frac{a}{J_{xx}}$, $b_2 = \frac{a}{J_{yy}}$, $b_3 = \frac{a}{J_{zz}}$, $b_4 = \frac{\cos x_1 \cos x_3}{m}$, $b_5 = \frac{u_x}{m}$, $b_6 = \frac{u_y}{m}$, $u_x = \cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi$, $u_y = \cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi$, d_i denotes external disturbances, f_i denotes actuator additive faults.

Assumption 1: It is supposed that D_i is bounded and changes slowly, that is $\dot{D}_i = 0$.

The control goals is to design the NFTSMFTC strategy to achieve $x, y, z, \phi, \theta, \psi$ tracking their reference commands $x_d, y_d, z_d, \phi_d, \theta_d, \psi_d$ under external disturbances and actuator additive faults, and ensure the asymptotic stability of the tracking error.

3. CONTROL SYSTEM DESIGN

In this section, the controllers of attitude subsystem and position subsystem are designed.

3.1. Design of attitude subsystem controller

The attitude subsystem (2) is divided into three second-order subsystems, which are roll angle subsystem (x_1, x_2) , pitch angle subsystem (x_3, x_4) and yaw angle subsystem (x_5, x_6) . For the dynamics of x_1 and x_2 , a NFTSM surface is given in the first step, and then the control input u_2 is designed in the second step, where the DO is employed to estimate D_1 . u_3, u_4 are designed in the same way.

The roll angle subsystem ϕ is

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= a_1 x_2 + b_1 u_2 + D_1,\end{aligned}\quad (4)$$

the tracking error is defined as $e_1 = x_1 - x_{1d}$.

Inspired by [26], the NFTSM surface is designed as

$$s_1 = e_1 + \frac{1}{\alpha_1} e_1^{\frac{p_1}{q_1}} + \frac{1}{\beta_1} \dot{e}_1^{\frac{p_2}{q_2}}. \quad (5)$$

The time derivative of (5) is

$$\begin{aligned}\dot{s}_1 &= \dot{e}_1 + \frac{p_1}{\alpha_1 q_1} e_1^{\frac{p_1}{q_1}-1} \dot{e}_1 + \frac{p_2}{\beta_1 q_2} \dot{e}_1^{\frac{p_2}{q_2}-1} (\dot{x}_2 - \ddot{x}_{1d}) \\ &= \dot{e}_1 + \frac{p_1}{\alpha_1 q_1} e_1^{\frac{p_1}{q_1}-1} \dot{e}_1 + M_1 (a_1 x_2 + b_1 u_2 + D_1 - \ddot{x}_{1d}),\end{aligned}\quad (6)$$

where $\alpha_1, \beta_1 > 0$, $M_1 = \frac{p_2}{\beta_1 q_2} \dot{e}_1^{\frac{p_2}{q_2}-1}$, p_1, p_2, q_1, q_2 are positive odd numbers, and $1 < \frac{p_2}{q_2} < 2$, $\frac{p_1}{q_1} > \frac{p_2}{q_2}$.

In (4), D_1 is estimated by following DO:

$$\begin{aligned}\hat{D}_1 &= k_{01}(x_2 - z_1), \\ \dot{z}_1 &= a_1 x_2 + b_1 u_2 + \hat{D}_1 - \frac{1}{k_{01}} M_1 s_1,\end{aligned}\quad (7)$$

where $k_{01} > \frac{1}{2}$, z_1 is an introduced auxiliary variable, $\hat{D}_1$ is estimation of D_1 .

From (6), control input u_2 is designed as

$$\begin{aligned}u_2 &= -\frac{1}{b_1} [a_1 x_2 + \hat{D}_1 - \ddot{x}_{1d} + k_1 s_1 + k_2 \text{sign}(s_1) \\ &\quad + \beta_1 \frac{q_2}{p_2} \dot{e}_1^{2-\frac{p_2}{q_2}} (1 + \frac{p_1}{\alpha_1 q_1} e_1^{\frac{p_1}{q_1}-1})],\end{aligned}\quad (8)$$

where $k_1, k_2 > 0$.

Define estimation error $\tilde{D}_1 = D_1 - \hat{D}_1$, based on (7) and (8), we have

$$\dot{\tilde{D}}_1 = k_{01} \tilde{D}_1 + M_1 s_1. \quad (9)$$

According to (8), equation (6) can be expressed as

$$\dot{s}_1 = M_1 [\tilde{D}_1 - k_1 s_1 - k_2 \text{sign}(s_1)]. \quad (10)$$

Remark 1: Compared with the traditional terminal SMC [27], the sliding mode surface (5) in this paper converges faster. In fact, sliding mode surface (5) contains $\frac{1}{\alpha_1} e_1^{\frac{p_1}{q_1}}$, (a) when error e_1 deviates from the equilibrium origin, $\frac{1}{\alpha_1} e_1^{\frac{p_1}{q_1}}$ works, and $\dot{e}_1 = -\frac{\beta_1}{\alpha_1} e_1^{\frac{h_1}{h_2}}$, ($h_1 = \frac{p_1}{q_1}, h_2 = \frac{p_2}{q_2}$), it can ensure faster convergence rate, (b) when the error e_1 is closed to the equilibrium point, $\dot{e}_1 = -\frac{1}{\beta_1} e_1^{\frac{p_2}{q_2}}$, it can also guarantee faster convergence rate. Moreover, compared with integral SMC [28], the chattering problem is effectively weakened.

Remark 2: In this paper, actuator faults and external disturbances are simultaneously tackled by the designed DO. Compared with [29], the computational complexity is reduced.

Theorem 1: According to the DO (7) and the controller (8), the tracking error e_1 of the roll angle subsystem (4) is asymptotically stable.

Proof: Choose the Lyapunov function $V_1 = \frac{1}{2} s_1^2 + \frac{1}{2} \tilde{D}_1^2$. Based on (9), (10) and Assumption 1, the time derivative of V_1 is

$$\begin{aligned}\dot{V}_1 &= s_1 [M_1 (\tilde{D}_1 - k_1 s_1 - k_2 \text{sign}(s_1))] + \tilde{D}_1 \dot{\tilde{D}}_1 - \tilde{D}_1 \dot{\tilde{D}}_1 \\ &\leq M_1 (s_1 \tilde{D}_1 - k_1 s_1^2 - k_2 |s_1|^2) + \frac{1}{2} \tilde{D}_1^2 - k_{01} \tilde{D}_1^2 - s_1 M_1 \tilde{D}_1 \\ &\leq -k_1 M_1 s_1^2 - k_2 M_1 |s_1| - (k_{01} - \frac{1}{2}) \tilde{D}_1^2,\end{aligned}\quad (11)$$

when $\dot{e}_1 \neq 0$, because of $k_1, k_2 > 0$, $M_1 > 0$, and when $k_{01} > \frac{1}{2}$, there is $\dot{V}_1 \leq 0$, when $\dot{e}_1 = 0$ and $e_1 \neq 0$, the system is unstable, so the sliding variable s_1 gradually reaches the sliding mode surface $s_1 = 0$ and remains on the sliding mode surface. In a word, the reachability of the sliding mode state can be proved. Later, the tracking error e_1 achieves asymptotic stability. $\square$

Similarly, the tracking errors are defined as $e_3 = x_3 - x_{3d}$, $e_5 = x_5 - x_{5d}$. The NFTSM surfaces are designed as $s_3 = e_3 + \frac{1}{\alpha_3} e_3^{\frac{p_3}{q_3}} + \frac{1}{\beta_3} \dot{e}_3^{\frac{p_4}{q_4}}$, $s_5 = e_5 + \frac{1}{\alpha_5} e_5^{\frac{p_5}{q_5}} + \frac{1}{\beta_5} \dot{e}_5^{\frac{p_6}{q_6}}$, where $\alpha_3, \beta_3, \alpha_5, \beta_5 > 0$, $p_j, q_j (j = 3, 4, 5, 6)$ are positive odd numbers and $1 < \frac{p_4}{q_4} < 2$, $\frac{p_3}{q_3} > \frac{p_4}{q_4}$, $1 < \frac{p_6}{q_6} < 2$, $\frac{p_5}{q_5} > \frac{p_6}{q_6}$, the DOs (12) and (13) are designed to deal with D_2 and D_3 of attitude subsystem (2).

$$\begin{aligned}\hat{D}_2 &= k_{02}(x_4 - z_2), \\ \dot{z}_2 &= a_2 x_4 + b_2 u_3 + \hat{D}_2 - \frac{1}{k_{02}} M_2 s_3,\end{aligned}\quad (12)$$

$$\begin{aligned}\hat{D}_3 &= k_{03}(x_6 - z_3), \\ \dot{z}_3 &= a_3 x_6 + b_3 u_4 + \hat{D}_3 - \frac{1}{k_{03}} M_3 s_5,\end{aligned}\quad (13)$$

where z_2, z_3 are auxiliary variables, $k_{02}, k_{03} > \frac{1}{2}$, $M_2 = \frac{p_4}{\beta_3 q_4} \dot{e}_3^{\frac{p_4}{q_4}-1}$, $M_3 = \frac{p_6}{\beta_5 q_6} \dot{e}_5^{\frac{p_6}{q_6}-1}$.

The control inputs u_3, u_4 are designed as

$$u_3 = -\frac{1}{b_2}[a_2x_4 + \hat{D}_2 - \ddot{x}_{3d} + k_3s_3 + k_4\text{sign}(s_3) + \beta_3\frac{q_3}{p_3}\dot{e}_3^{2-\frac{p_4}{q_4}}(1 + \frac{p_3}{\alpha_3q_3}e_3^{\frac{p_3}{q_3}-1})], \quad (14)$$

$$u_4 = -\frac{1}{b_3}[a_3x_6 + \hat{D}_3 - \ddot{x}_{5d} + k_5s_5 + k_6\text{sign}(s_5) + \beta_5\frac{q_5}{p_5}\dot{e}_5^{2-\frac{p_6}{q_6}}(1 + \frac{p_5}{\alpha_5q_5}e_5^{\frac{p_5}{q_5}-1})], \quad (15)$$

where $k_j > 0$, $\hat{D}_2$ and $\hat{D}_3$ are estimations of D_2 and D_3 .

3.2. Design of position subsystem controller

The position subsystem (3) is divided into three second-order subsystems, and the controllers u_1 , u_x and u_y are designed, respectively. For the dynamics of x_7 and x_8 , a NFTSM surface is developed in the first step, and then the control input u_1 is designed in the second step, where the DO is employed to estimate D_4 . u_x, u_y are designed in the same way.

The altitude subsystem z is

$$\begin{aligned} \dot{x}_7 &= x_8, \\ \dot{x}_8 &= a_4x_8 + \frac{\cos x_1 \cos x_3}{m}u_1 - g + D_4, \end{aligned} \quad (16)$$

the tracking error is defined as $e_7 = x_7 - x_{7d}$.

Inspired by [26], a NFTSM surface is designed

$$s_7 = e_7 + \frac{1}{\alpha_7}e_7^{\frac{p_7}{q_7}} + \frac{1}{\beta_7}\dot{e}_7^{\frac{p_8}{q_8}}. \quad (17)$$

Similar to (6), the time derivative of (17) is

$$\begin{aligned} \dot{s}_7 &= \dot{e}_7 + \frac{p_7}{\alpha_7q_7}e_7^{\frac{p_7}{q_7}-1}\dot{e}_7 + M_4(a_4x_8 \\ &+ \frac{\cos x_1 \cos x_3}{m}u_1 + D_4 - g - \ddot{x}_{7d}), \end{aligned} \quad (18)$$

where $\alpha_7, \beta_7 > 0$, $M_4 = \frac{p_8}{\beta_7q_8}\dot{e}_7^{\frac{p_8}{q_8}-1}$, p_7, p_8, q_7, q_8 are positive odd numbers and $1 < \frac{p_8}{q_8} < 2$, $\frac{p_7}{q_7} > \frac{p_8}{q_8}$.

The DO is designed as follows:

$$\begin{aligned} \hat{D}_4 &= k_{04}(x_8 - z_4), \\ \dot{z}_4 &= a_4x_8 + \frac{\cos x_1 \cos x_3}{m}u_1 - g + \hat{D}_4 - \frac{1}{k_{04}}M_2s_7, \end{aligned} \quad (19)$$

where $k_{04} > \frac{1}{2}$, z_4 is an introduced auxiliary variable, $\hat{D}_4$ is estimation of D_4 .

Based on (18), the control input u_1 is designed as

$$\begin{aligned} u_1 &= -\frac{m}{\cos x_1 \cos x_3}[\beta_7\frac{q_8}{p_8}\dot{e}_7^{2-\frac{p_8}{q_8}}(1 + \frac{p_7}{\alpha_7q_7}e_7^{\frac{p_7}{q_7}-1}) \\ &+ a_4x_8 + \hat{D}_4 - \ddot{x}_{7d} - g + k_7s_7 + k_8\text{sign}(s_7)], \end{aligned} \quad (20)$$

where $k_7, k_8 > 0$.

Define estimation error $\tilde{D}_4 = D_4 - \hat{D}_4$, based on (19) and (20), we have

$$\dot{\tilde{D}}_4 = k_{04}\tilde{D}_4 + M_4s_7. \quad (21)$$

According to (20), equation (18) can be expressed as

$$\dot{s}_7 = M_4[\tilde{D}_4 - k_7s_7 - k_8\text{sign}(s_7)]. \quad (22)$$

Theorem 2: Based on the controller (19) and the DO (20), the tracking error e_7 of the altitude subsystem (19) is asymptotically stable.

Proof: Similar to Theorem 1, the Lyapunov function is $V_2 = \frac{1}{2}s_7^2 + \frac{1}{2}\tilde{D}_4^2$. From (21), (22) and Assumption 1, the time derivative of V_2 is $\dot{V}_2 \leq -k_7M_4s_7^2 - k_8M_4|s_7| - (k_{04} - \frac{1}{2})\tilde{D}_4^2$. Therefore, the tracking error e_7 achieves asymptotic stability. $\square$

Similarly, the tracking errors are defined as $e_9 = x_9 - x_{9d}$, $e_{11} = x_{11} - x_{11d}$. The NFTSM surfaces are designed as $s_9 = e_9 + \frac{1}{\alpha_9}e_9^{\frac{p_9}{q_9}} + \frac{1}{\beta_9}\dot{e}_9^{\frac{p_{10}}{q_{10}}}$, $s_{11} = e_{11} + \frac{1}{\alpha_{11}}e_{11}^{\frac{p_{11}}{q_{11}}} + \frac{1}{\beta_{11}}\dot{e}_{11}^{\frac{p_{12}}{q_{12}}}$, where $\alpha_9, \beta_9, \alpha_{11}, \beta_{11} > 0$, p_l and q_l ($l = 9, 10, 11, 12$) are positive odd numbers and $1 < \frac{p_{10}}{q_{10}} < 2$, $\frac{p_9}{q_9} > \frac{p_{10}}{q_{10}}$, $\frac{p_{11}}{q_{11}} > \frac{p_{12}}{q_{12}}$, $1 < \frac{p_{12}}{q_{12}} < 2$. the DOs (23) and (24) are designed to deal with D_5 and D_6 of position subsystem (4).

$$\begin{aligned} \hat{D}_5 &= k_{05}(x_{10} - z_5), \\ \dot{z}_5 &= a_5x_{10} + \frac{\cos x_1 \cos x_3}{m}u_x u_1 + \hat{D}_5 - \frac{1}{k_{05}}M_5s_9, \end{aligned} \quad (23)$$

$$\begin{aligned} \hat{D}_6 &= k_{06}(x_{12} - z_6), \\ \dot{z}_6 &= a_6x_{12} + \frac{\cos x_1 \cos x_3}{m}u_y u_1 + \hat{D}_6 - \frac{1}{k_{06}}M_6s_{11}, \end{aligned} \quad (24)$$

where z_5, z_6 are auxiliary variables, $k_{05}, k_{06} > \frac{1}{2}$, $M_5 = \frac{p_{10}}{\beta_9q_{10}}\dot{e}_9^{\frac{p_{10}}{q_{10}}-1}$, $M_6 = \frac{p_{12}}{\beta_{11}q_{12}}\dot{e}_{11}^{\frac{p_{12}}{q_{12}}-1}$.

The control inputs are designed as

$$\begin{aligned} u_x &= -\frac{m}{u_1}[\beta_9\frac{q_{10}}{p_{10}}\dot{e}_9^{2-\frac{q_{10}}{p_{10}}}(1 + \frac{p_9}{\alpha_9q_9}e_9^{\frac{q_9}{p_9}-1}) \\ &+ a_5x_{10} + \hat{D}_5 - \ddot{x}_{9d} + k_9s_9 + k_{10}\text{sign}(s_9)], \end{aligned} \quad (25)$$

$$\begin{aligned} u_y &= -\frac{m}{u_1}[\beta_{11}\frac{q_{12}}{p_{12}}\dot{e}_{11}^{2-\frac{q_{12}}{p_{12}}}(1 + \frac{p_{11}}{\alpha_{11}q_{11}}e_{11}^{\frac{q_{11}}{p_{11}}-1}) \\ &+ a_6x_{12} + \hat{D}_6 - \ddot{x}_{11d} + k_{11}s_{11} + k_{12}\text{sign}(s_{11})], \end{aligned} \quad (26)$$

where $k_l > 0$, $\hat{D}_5$ and $\hat{D}_6$ are estimations of D_5 and D_6 .

Remark 3: In the control inputs (8), (14), (15), (20), (25) and (26), the chattering problem may be caused by sign function. To avoid this problem, the following continuous function is applied to replace the sign function,

$$\phi(\rho, s_\kappa) = \frac{e^{\rho s_\kappa} - 1}{e^{\rho s_\kappa} + 1}, \quad (\rho > 1, \kappa = 1, 3, 5, 7, 9, 11).$$

3.3. Inverse solution of desired values of roll angle and pitch angle

The system of a quadrotor UAV is under-actuated, it is necessary to convert it a fully actuated system. The control inputs u_x, u_y are designed in (25) and (26). From (1), the position and attitude influence each other, according to the controller design of position and attitude subsystems before, it can be obtained as

$$\begin{aligned} u_x &= \cos \phi_d \sin \theta_d \cos \psi_d + \sin \phi_d \sin \psi_d, \\ u_y &= \cos \phi_d \sin \theta_d \sin \psi_d - \sin \phi_d \cos \psi_d. \end{aligned} \quad (27)$$

According to (25), (26) and (27), we obtain

$$\begin{aligned} \phi_d &= \arcsin(u_x \sin \psi_d - u_y \cos \psi_d), \\ \theta_d &= \arcsin\left(\frac{u_x \cos \psi_d + u_y \sin \psi_d}{\cos \phi_d}\right). \end{aligned} \quad (28)$$

4. SIMULATION

In this section, simulations are shown to verify the effectiveness of the proposed control scheme. The initial conditions of the quadrotor UAV are given as $(x, y, z) = (0.97, 0.95, 0)$ m, $(\phi, \theta, \psi) = (0, 0, 0)$ rad. The reference commands are $(x_d, y_d, z_d) = (1, 1, 1)$ m, $\psi_d = 0$ rad. The external disturbance is $d_i = 0.2 \sin(2\pi/25t)$, the actuator additive fault occurs 10 seconds later and the fault is $f_i = 0.5 + 14 \sin(t - 10)$. The parameters of quadrotor UAV are given in Table 1 and the controller parameters are shown in Table 2. The parameters of the DOs are $k_{01} = 0.7$, $k_{02} = k_{03} = 0.9$, $k_{04} = 1.5$, $k_{05} = k_{06} = 0.8$.

Table 1. Parameters of a quadrotor UAV.

Name	Parameter	Value	Unit
Mass	m	2	kg
Arm length	a	0.2	m
Acceleration	g	9.8	kg/m ²
Inertia around x axis	J_{xx}	1.25	kg · m ²
Inertia around y axis	J_{yy}	1.25	kg · m ²
Inertia around z axis	J_{zz}	2.5	kg · m ²
Drag coefficients	$G_1 - G_3$	0.01	kg/s
Drag coefficients	$G_4 - G_6$	0.01	kg/rad

Table 2. Controller parameters.

Variable	Value	Variable	Value	Variable	Value
q_1, q_3, q_5	5	α_9, α_{11}	5	k_6, k_7, k_8	2
p_1, p_3, p_5	7	$\beta_3, \beta_5, \beta_7$	2	k_{10}	2000
k_9, k_{12}	2	β_9, β_{11}	18	k_{11}	50
p_2, p_4, p_6	9	k_1, k_2	0.001	p_8, p_{10}, p_{12}	13
α_1	5	k_3	0.0009	q_8	9
β_1	325	k_4	0.2	q_{10}, q_{12}	11
$\alpha_3, \alpha_5, \alpha_7$	2	k_5	4	q_2, q_4, q_6	7
p_7, p_9, p_{11}	11	q_7	7	q_9, q_{11}	9

4.1. The efficiency of NFTSMC under different factors

In the simulation, the following two cases are considered.

Case 1: Only external disturbance is considered.

Case 2: Both external disturbance and actuator additive fault are considered.

Simulation results cases are shown in Figs. 1-3.

Position tracking performance is described in Fig. 1. It can be seen from Fig. 1(a) that the position x and the position y can track their reference commands within 1 second in both two cases. In Case 2, although there is a small fluctuation when actuator fault occurs in the 10th second, it is

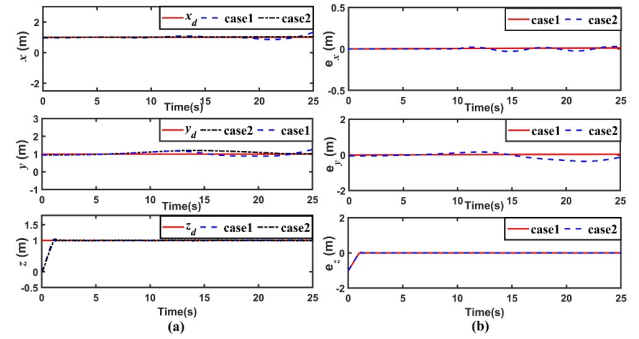


Fig. 1. Time response of position tracking and tracking errors of Case 1 and Case 2.

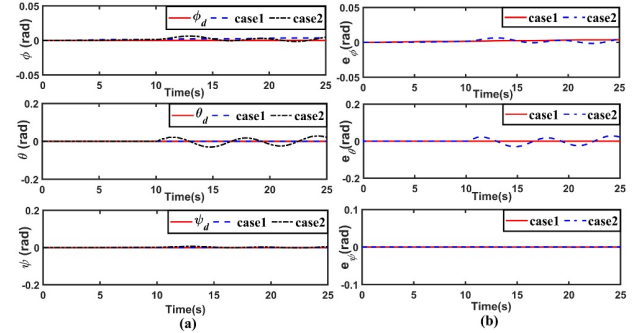


Fig. 2. Time response of attitude tracking and tracking errors of Case 1 and Case 2.

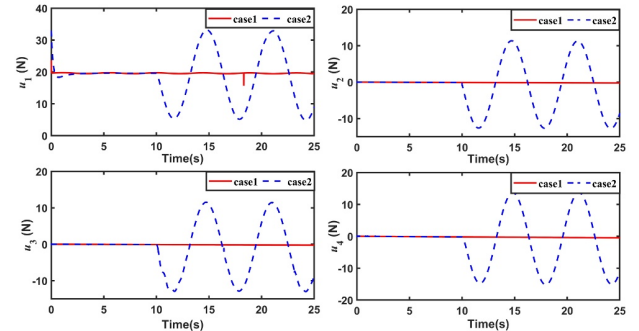


Fig. 3. Control inputs of Case 1 and Case 2.

still within the controllable range. In both cases, the position z is stable within 5 seconds. It can be seen from Fig. 1(b) that the tracking performance of x and y is good in both cases, and the tracking accuracy of Case 1 is better than that of Case 2.

Attitude tracking performance is shown as Fig. 2. In Fig. 2(a), roll angle ϕ maintains at -0.05 rad to 0.05 rad, pitch angle θ and yaw angle ψ keep at -0.2 rad to 0.2 rad. It can be seen from Fig. 2(b) that the tracking error of ϕ and θ in Case 2 have slight fluctuations after 10 s, but it does not affect the tracking performance. The accuracy of Case 1 is better than that of Case 2. Fig. 3 shows the time history of control inputs. The control input u_1 in Case 1 shows a short changing between 15 s and 20 s and it remains smooth after 20 s. Since the fault is selected as a sinusoidal time-varying function, control inputs of Case 2 have a sinusoidal change after 10 s, and the change trend is relatively smooth.

4.2. Compared simulations of different methods

To validate the superiority of the proposed control scheme, the following comparisons are done.

Case 3: The DO-based NFTSMC scheme is compared with the DO-based NTSMC scheme.

Case 4: The way handles actuator fault based on DO is compared with adaptive method [23].

Simulation results of Case 3 are shown in Figs. 4-6. Position tracking performances of two methods are shown in Fig. 4(a), and the tracking errors are given in Fig. 5(a). It can be seen from Fig. 4(a) that the tracking performance of position is good under two methods. It can be clearly seen from Fig. 5(a) that the tracking error convergence rate of DO-based NFTSMC method is faster than NTSMC.

Attitude tracking performance of two methods is shown in Fig. 4(b), and Fig. 5(b) shows the attitude tracking error. In Fig. 4(b), attitude angle ϕ , θ , ψ maintain at -0.05 rad to 0.05 rad. Actuator additive fault occurs from the 10th second, although the attitude tracking errors e_ϕ , e_θ , e_ψ can be kept at a small range under two methods, it can be clearly obtained from e_ψ that there is a large error under the DO-based NTSMC method after the actuator fault occurs. In Fig. 6, the time response of control inputs is provided.

Simulation results of Case 4 are given in Figs. 7-9. It can be observed from Fig. 7 that the designed two control strategies achieve the stable tracking of position and attitude. Position tracking performance of the DO-based fault-tolerant control method is better than that of the adaptive method. It is clear from Fig. 8 that when the actuator fault occurs from the 10th second, the tracking errors become big under the fault-tolerant control strategy of adaptive method. In Fig. 9, the time history of control inputs is provided. The control inputs have slightly chattering phenomena when the additional actuator faults are handled by the adaptive method.

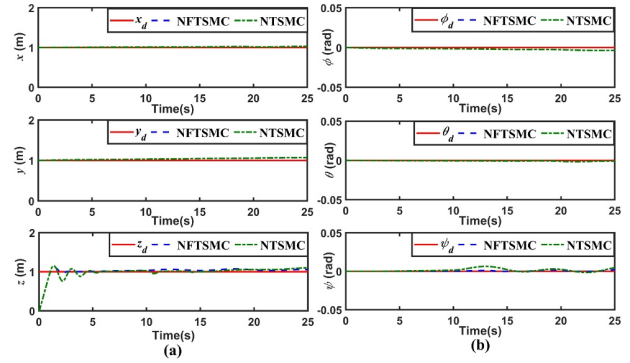


Fig. 4. Time response of position and attitude tracking of Case 3.

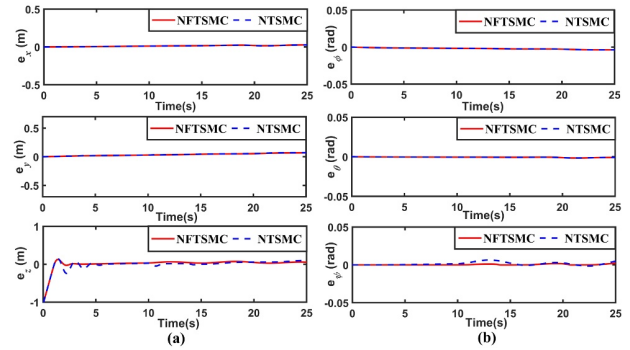


Fig. 5. Time response of position and attitude tracking errors of Case 3.

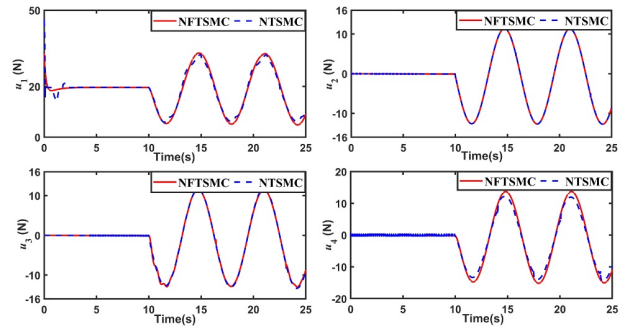


Fig. 6. Time response of control inputs of Case 3.

4.3. Several waypoint tracking trajectories

To verify the effectiveness of the proposed control method better, the following several waypoints tracking trajectories are considered.

$$x_d = 10 \sin\left(\frac{2\pi t}{50}\right), y_d = 10 \cos\left(\frac{2\pi t}{50}\right), z_d = \frac{1}{6}t.$$

As shown in Figs. 10-12, it is clear that the designed controller can achieve the stable tracking of position and attitude with external disturbance and actuator fault.

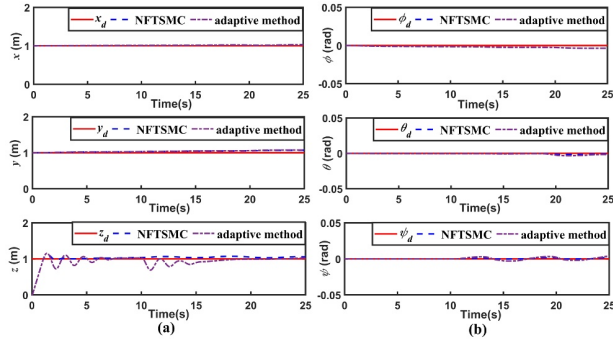


Fig. 7. Time response of position and attitude tracking of Case 4.

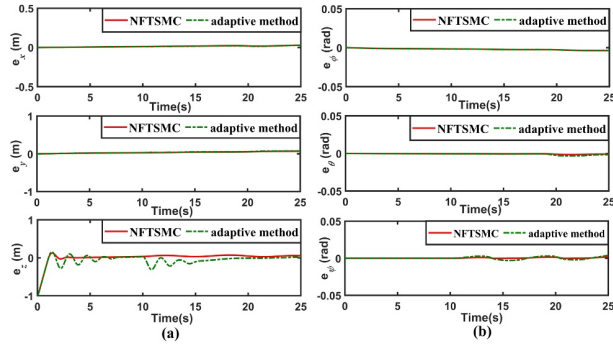


Fig. 8. Time response of position and attitude tracking errors of Case 4.

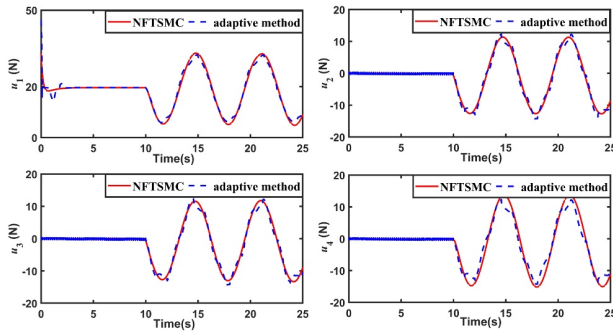


Fig. 9. Time response of control inputs of Case 4.

5. CONCLUSION

This paper proposes a nonsingular fast terminal sliding mode fault-tolerant control scheme for the tracking control problem of a quadrotor UAV under external disturbance and actuator additive fault. Combined with DO and NFTSMC, a nonsingular fast terminal sliding mode fault-tolerant controller is designed. The DO is employed to handle external disturbance and actuator additive fault. Stability analysis shows that the designed control strategy achieves the asymptotic stability of the tracking error. Simulation results demonstrate that the effectiveness of

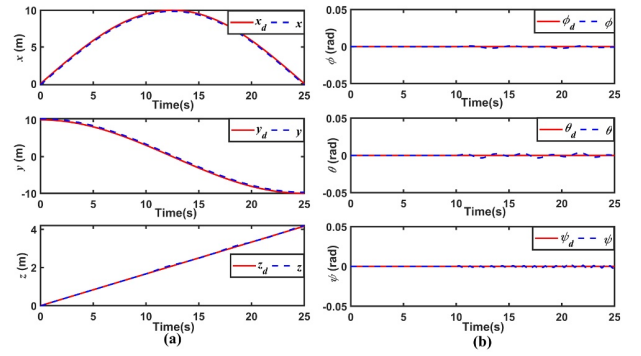


Fig. 10. Time response of position and attitude tracking.

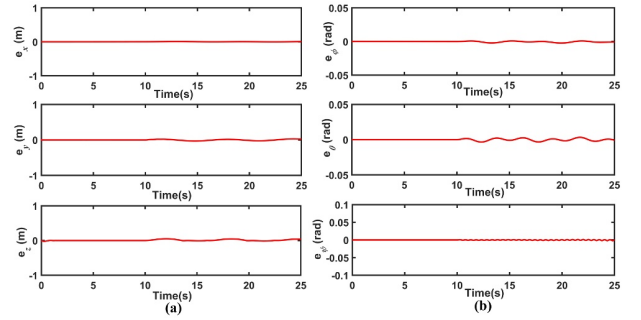


Fig. 11. Time response of position and attitude tracking errors.

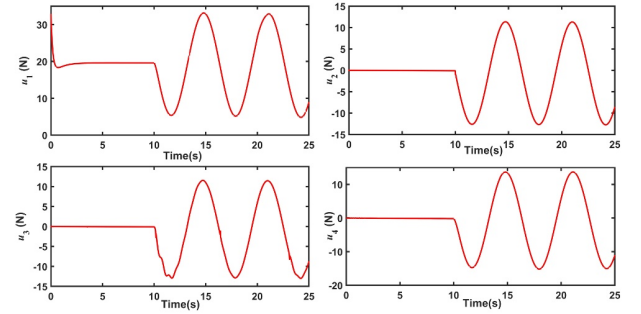


Fig. 12. Time response of control inputs.

the designed control scheme against external disturbance and actuator additive fault is achieved.

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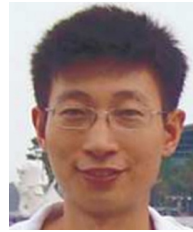


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